

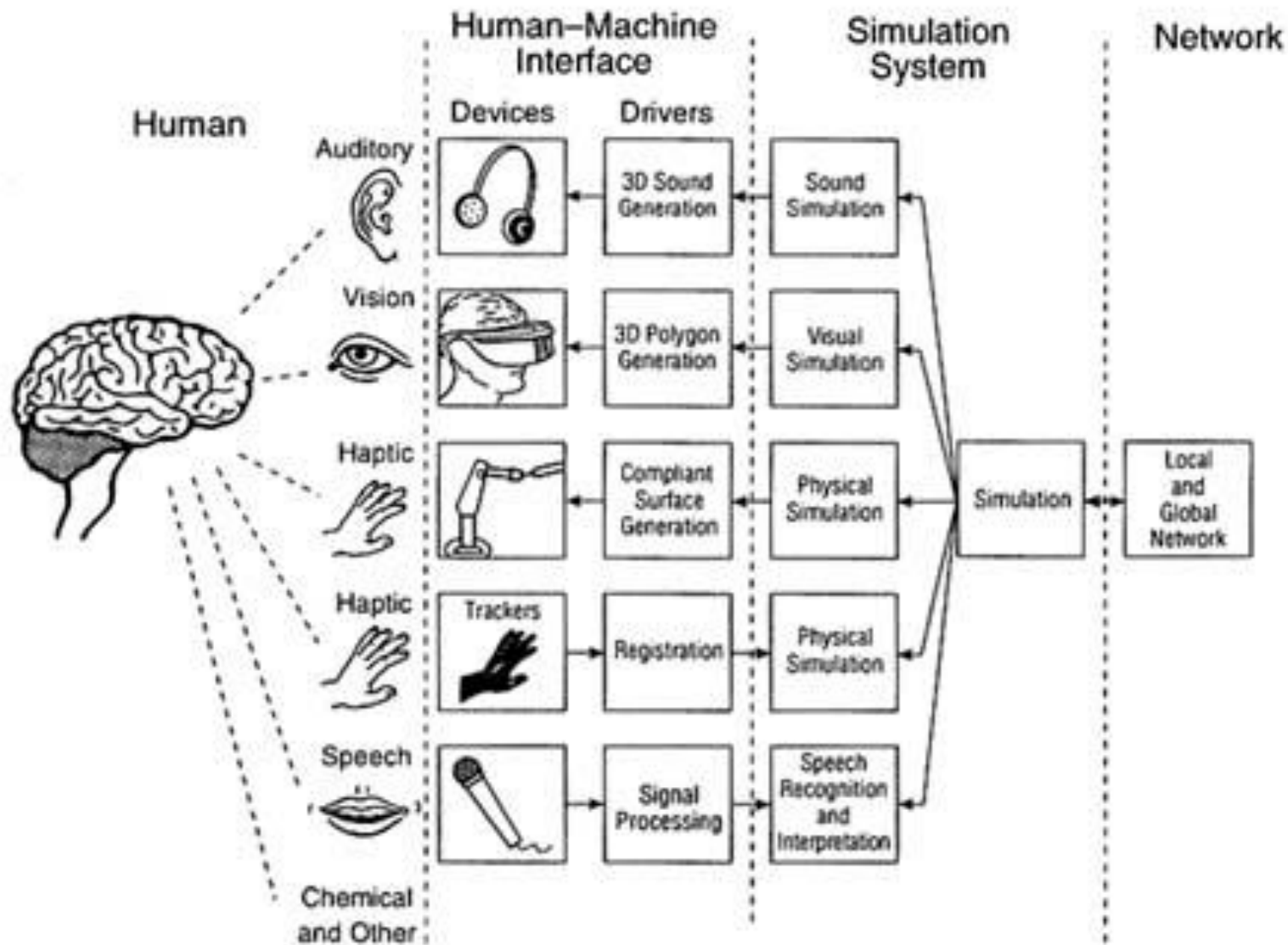


Human computer interaction

Human and Virtual Reality

Material from S.M. LaValle, Mark Billinghurst, Bruce Thomas

Virtual Reality





More on perception

- There are many issues about human perception that must be considered in depth when analyzing interaction in VR
- Are/May be neglected for classic HCI base on intentional actions and simple stimuli
 - Awareness, attention
 - Information fusion
 - Hierarchical processing
 - Adaptation
 - psychophysics

Senses?

Sensory System	Modality	Stimulus Energy	Receptor Class	Receptor Cell Types
Somatosensory	Touch	Tap, flutter 5–40 Hz	Cutaneous mechanoreceptor	Meissner corpuscles
Somatosensory	Touch	Motion	Cutaneous mechanoreceptor	Hair follicle receptors
Somatosensory	Touch	Deep pressure, vibration 60–300 Hz	Cutaneous mechanoreceptor	Pacinian corpuscles
Somatosensory	Touch	Touch, pressure	Cutaneous mechanoreceptor	Merkel cells
Somatosensory	Touch	Sustained pressure	Cutaneous mechanoreceptor	Ruffini corpuscles
Somatosensory	Proprioception	Stretch	Mechanoreceptor	Muscle spindles
Somatosensory	Proprioception	Tension	Mechanoreceptor	Golgi tendon organ
Somatosensory	Temperature	Thermal	Thermoreceptor	Cold and warm receptors
Somatosensory	Pain	Chemical, thermal, and mechanical	Chemoreceptor, thermoreceptor, and mechanoreceptor	Polymodal receptors or chemical, thermal, and mechanical nociceptors
Somatosensory	Itch	Chemical	Chemoreceptor	Chemical nociceptor
Visual	Vision	Light	Photoreceptor	Rods, cones
Auditory	Hearing	Sound	Mechanoreceptor	Hair cells (cochlea)
Vestibular	Balance	Angular acceleration	Mechanoreceptor	Hair cells (semicircular canals)
Vestibular	Balance	Linear acceleration, gravity	Mechanoreceptor	Hair cells (otolith organs)
Olfactory	Smell	Chemical	Chemoreceptor	Olfactory sensory neuron
Gustatory	Taste	Chemical	Chemoreceptor	Taste buds



More than 5 senses

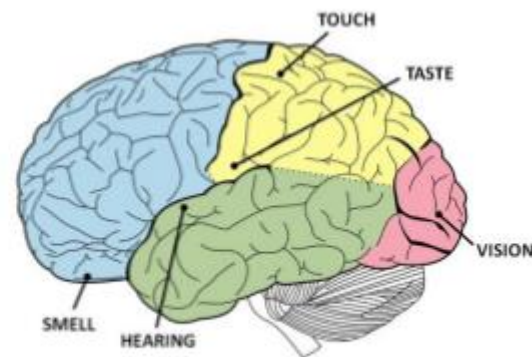
- Proprioception = sense of body position what is your body doing right now
- Equilibrium = balance
- Acceleration
- Nociception = sense of pain
- Temperature
- Satiety
- Thirst
- Micturition
- Amount of CO₂ and Na in blood



Senses

- Vision is the main channel.
- But other input are important and **must be coherent**
- And sensing is an **active process** that depends on previous inputs, **not in a conscious way**

- Percentage of neurons in brain devoted to each sense
 - Sight – 30%
 - Touch – 8%
 - Hearing – 2%
 - Smell - < 1%
- Over 60% of brain involved with vision in some way



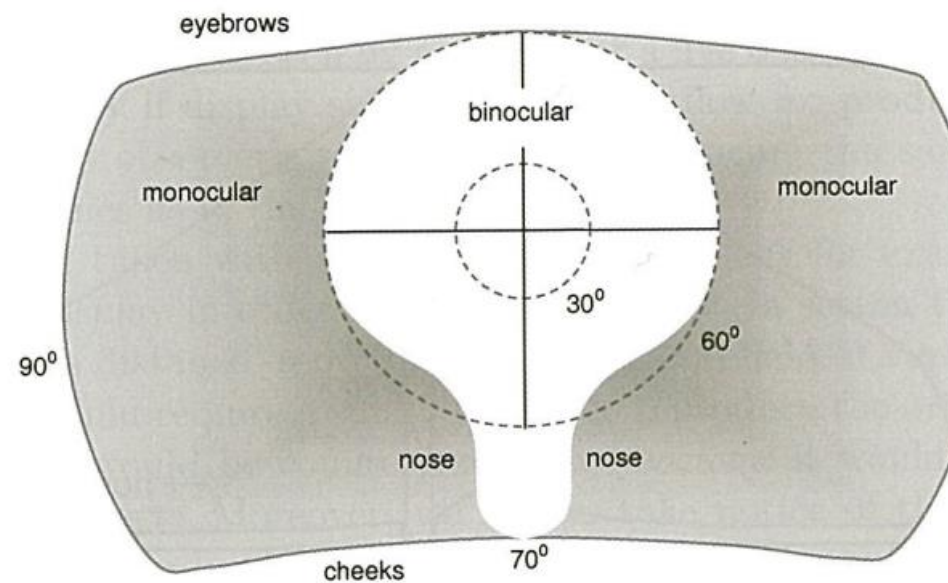
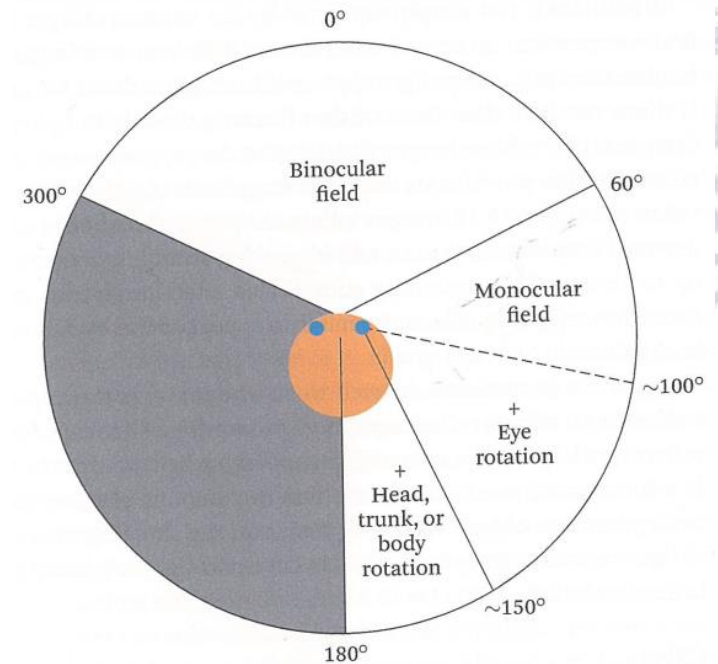


Adaptation

- Perceived effect of stimuli changes over time.
 - the perceived loudness of motor noise in an aircraft or car decreases within minutes.
 - The optical system of our eyes and the photoreceptor sensitivities adapt to change perceived brightness.
- Over long periods, perceptual training can lead to adaptation;
 - In military training simulations, sickness experienced by soldiers appears to be less than expected, perhaps due to regular exposure
 - The same seems to be true of experienced video game players. Those who have spent many hours and days in front of large screens playing first-person shooter games apparently experience less vection when locomoting themselves in VR.
- Adaptation therefore becomes a crucial factor for VR.

Constraints for VR systems

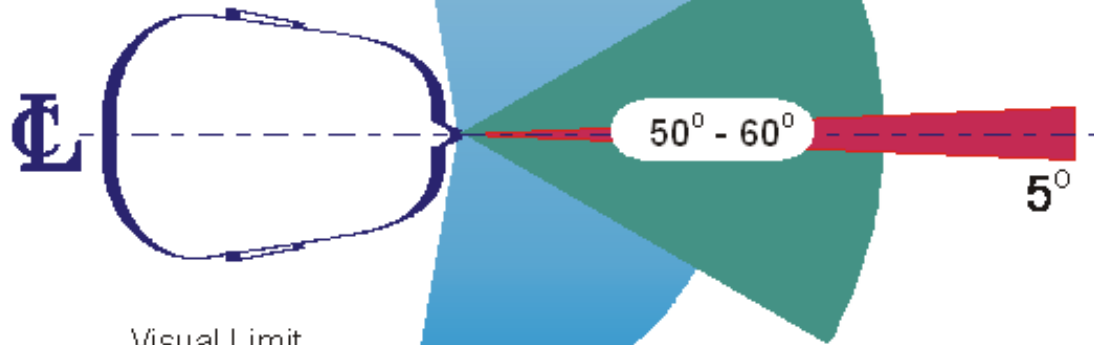
- Human frontal FOV
 - 100-110 horizontal per eye
 - Up to 200 -220 horizontal
 - Both eyes
 - 110 -120 stereo overlap
- Foveal Field
 - 60 horiz/vertically
 - Both eyes can see in focus





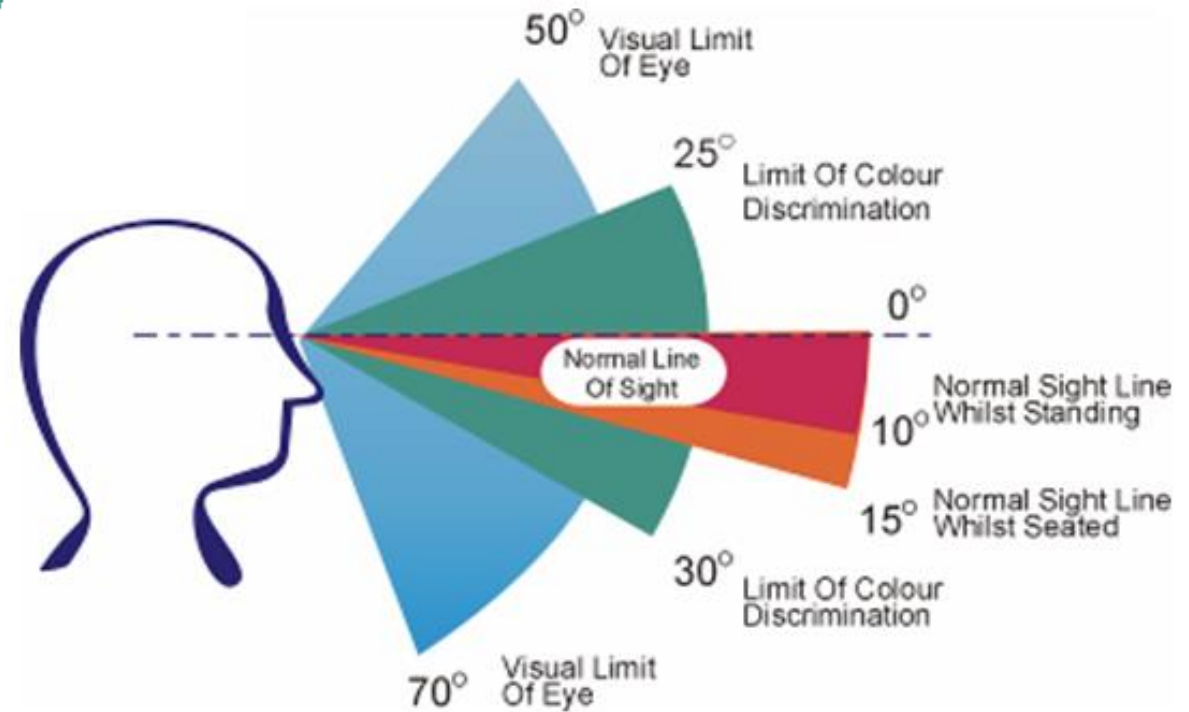
Visual Limit
Of Left Eye

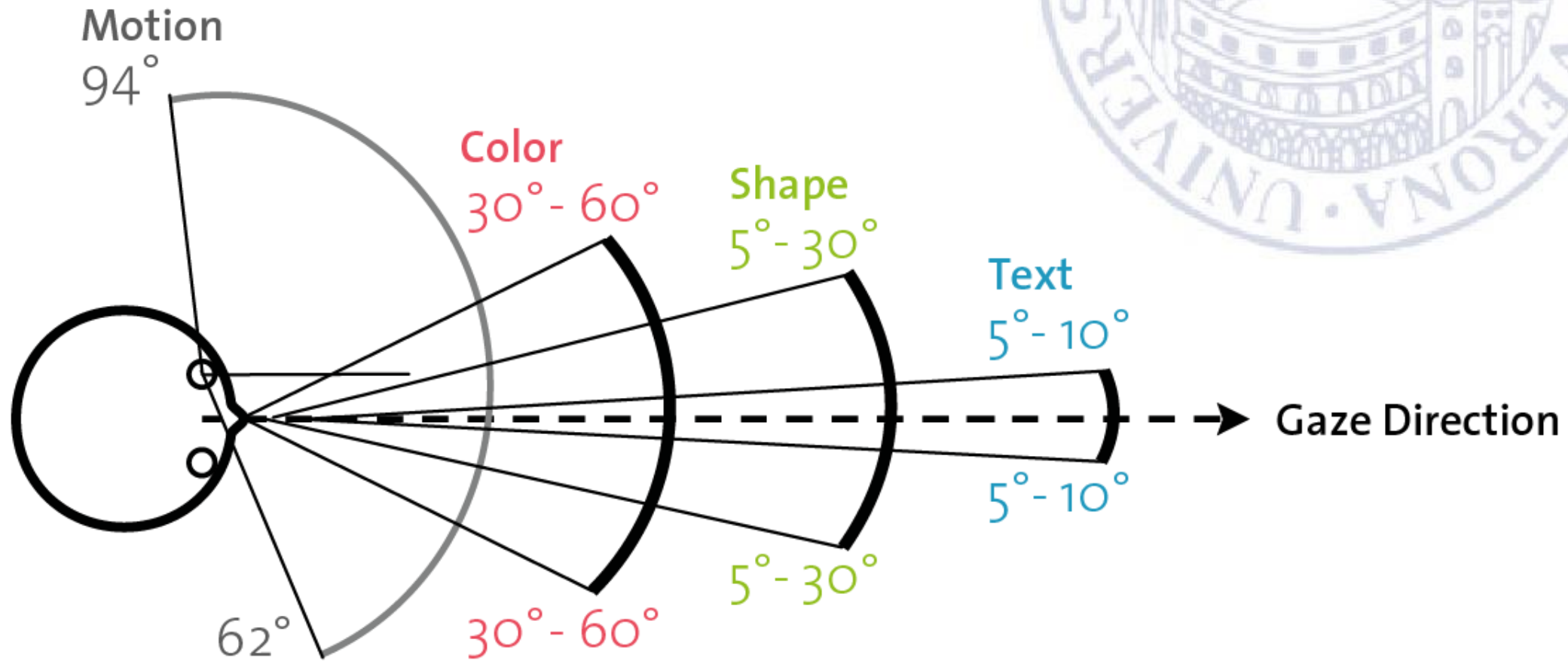
104° to 94°



Visual Limit
Of Right Eye

104° to 94°







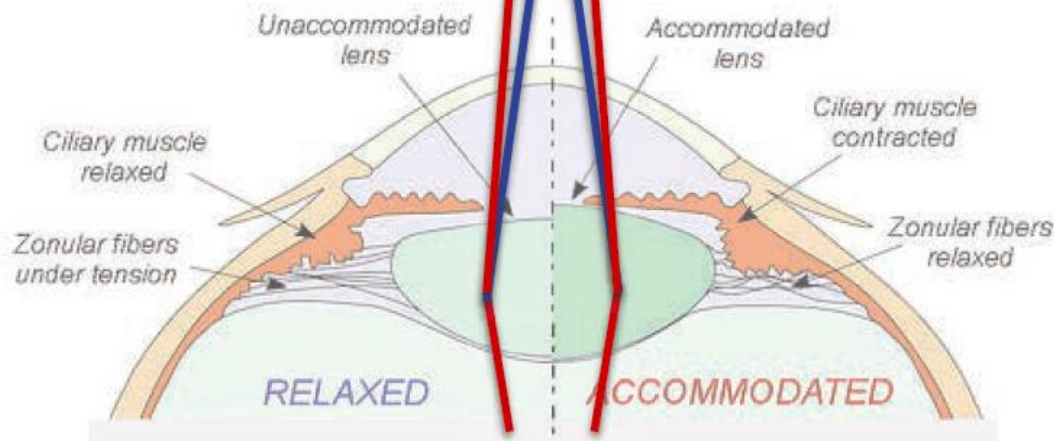
Oculomotor Processes

far focus

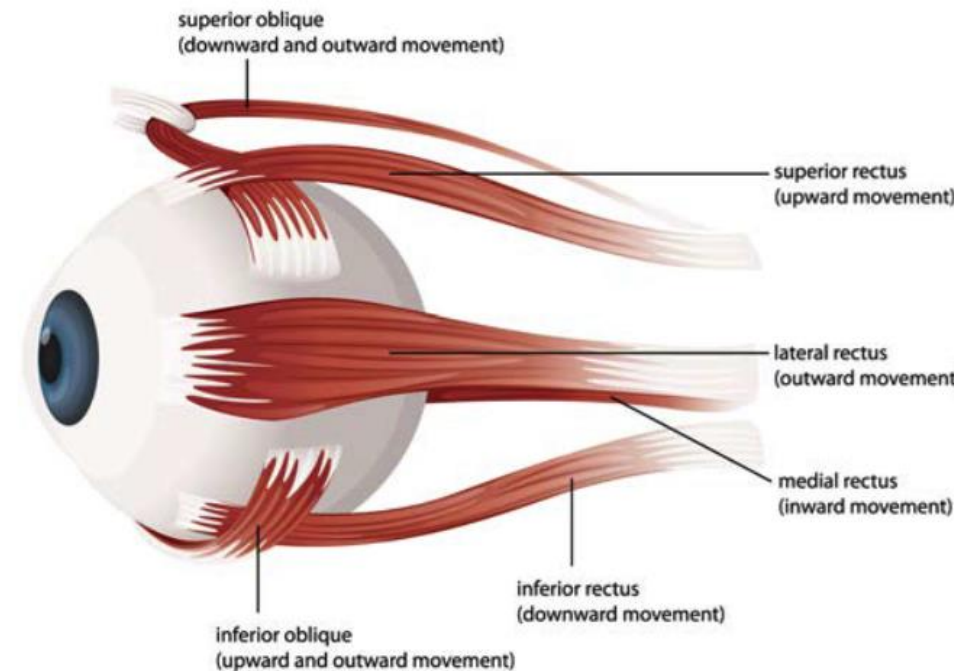
16 years: ~8cm to ∞

50 years: ~50cm to ∞ (mostly irrelevant)

near focus



adithyakiran.wordpress.com





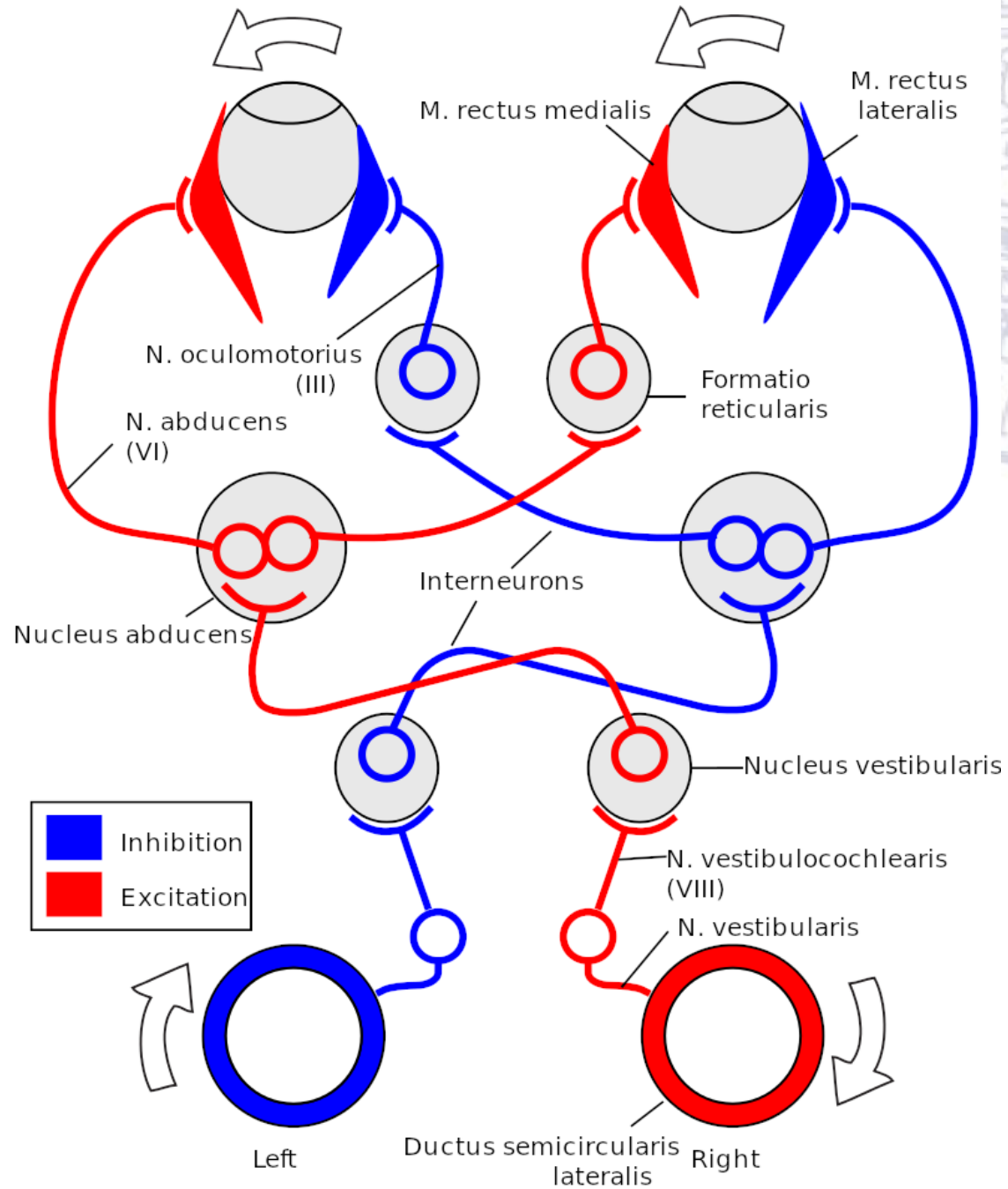
Many movements

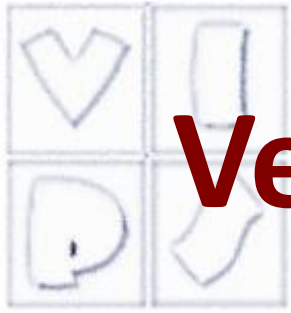
- **Saccades** less than 45ms with rotations of about 90° per second.
 - To quickly relocate the fovea so that important features in a scene are sensed with highest visual acuity.
 - illusion of high acuity over a large angular range.
 - no awareness of them, we have the ability to consciously control them as we choose features for fixation.
- **Vestibulo-ocular reflex (VOR)**
 - One of the most important motions to understand for VR
 - Involuntary. Try to avoid rotating your eyes while paying attention to your finger and rotating your head.
 - Based on angular accelerations sensed by vestibular organs, signals are sent to the eye muscles to provide the appropriate counter motion.
 - The main purpose of the VOR is to provide image stabilization



Many movements

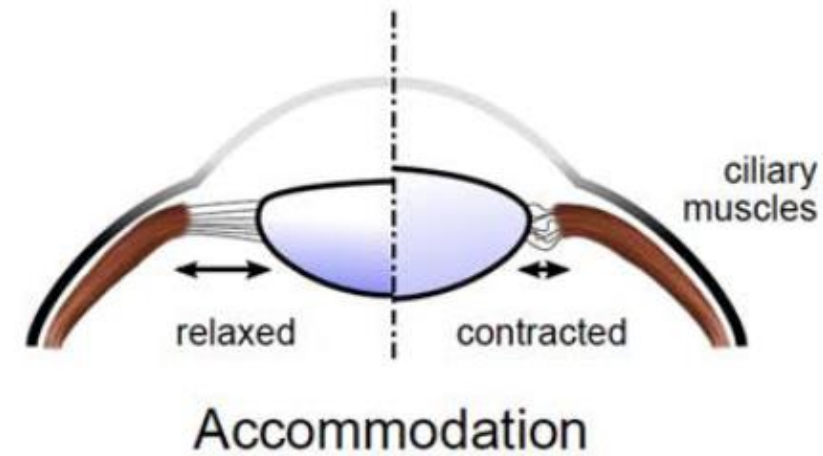
- **Smooth pursuit**
 - the eye slowly rotates to track a moving target feature. Examples are a car, a tennis ball, or a person walking by. Usually less than 30° per second, which is much slower than for saccades. Used to reduce motion blur on the retina;
- **Optokinetic reflex**
 - when a fast object speeds along. The eyes rapidly and involuntarily choose features for tracking on the object
- **Vergence Stereopsis**
 - convergence motion occurs. Eyes are rotating so that the pupils are becoming closer.
 - divergence motion occurs, pupils move further apart.
 - provide important information about distance of objects.
- **Microsaccades**
 - small, involuntary jerks of less than one degree that trace out an erratic path. They are believed to augment many other processes, including control of fixations, reduction of perceptual fading due to adaptation, improvement of visual acuity



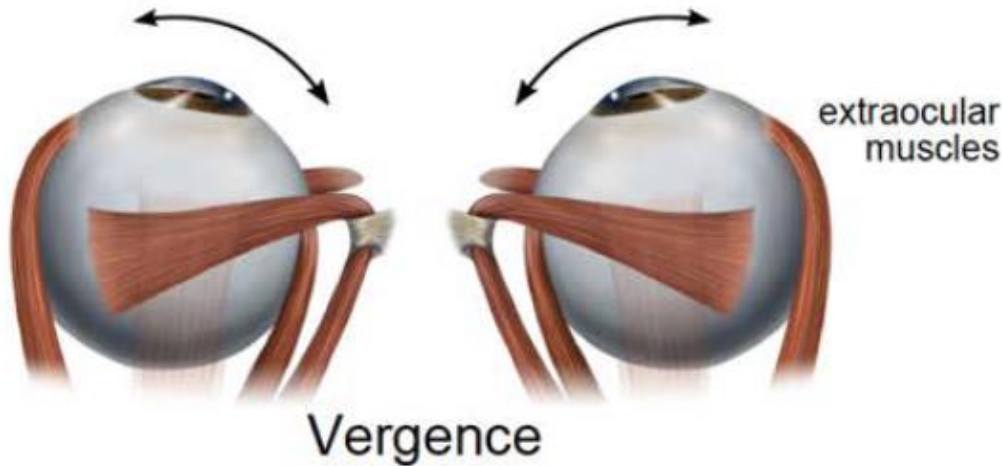


Vergence and accommodation

Focus Cues (Monocular)



Stereopsis (Binocular)



Binocular Disparity



Retinal Blur

Oculomotor Cue

Visual Cue

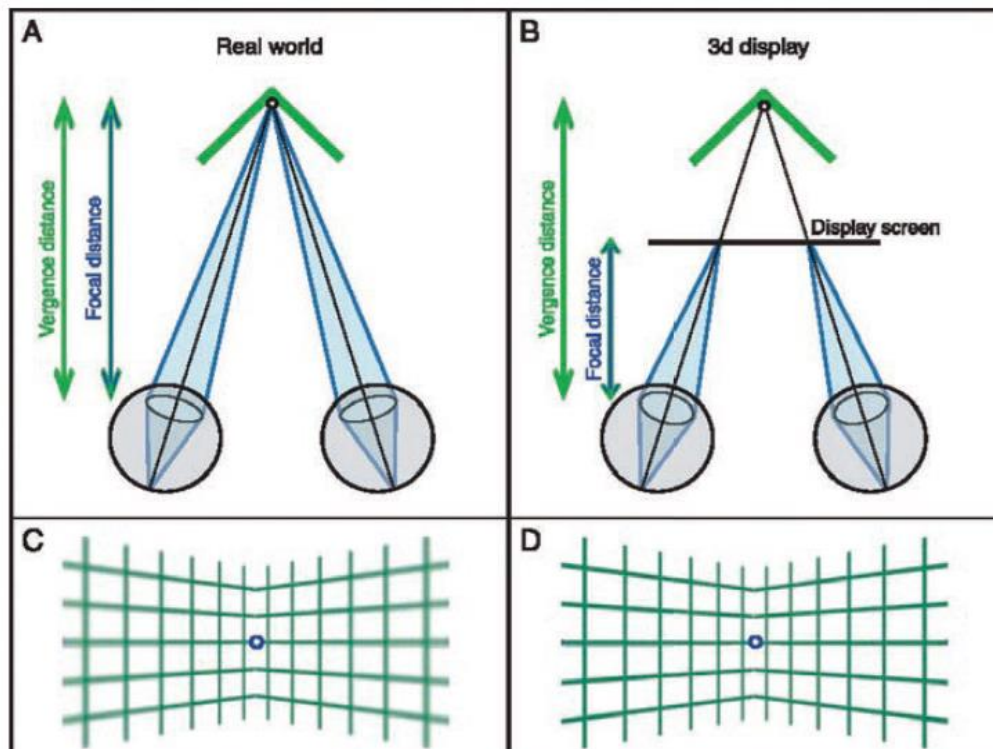


video



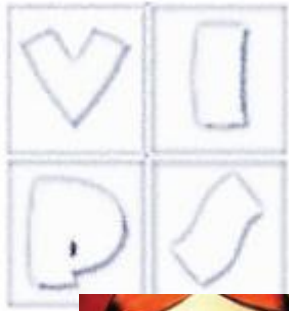
Can conflict

- Looking at real objects, vergence and focal distance match
 - In Virtual Reality, vergence and accommodation can mismatch
 - Focusing on HMD screen, but accommodating for virtual object behind screen



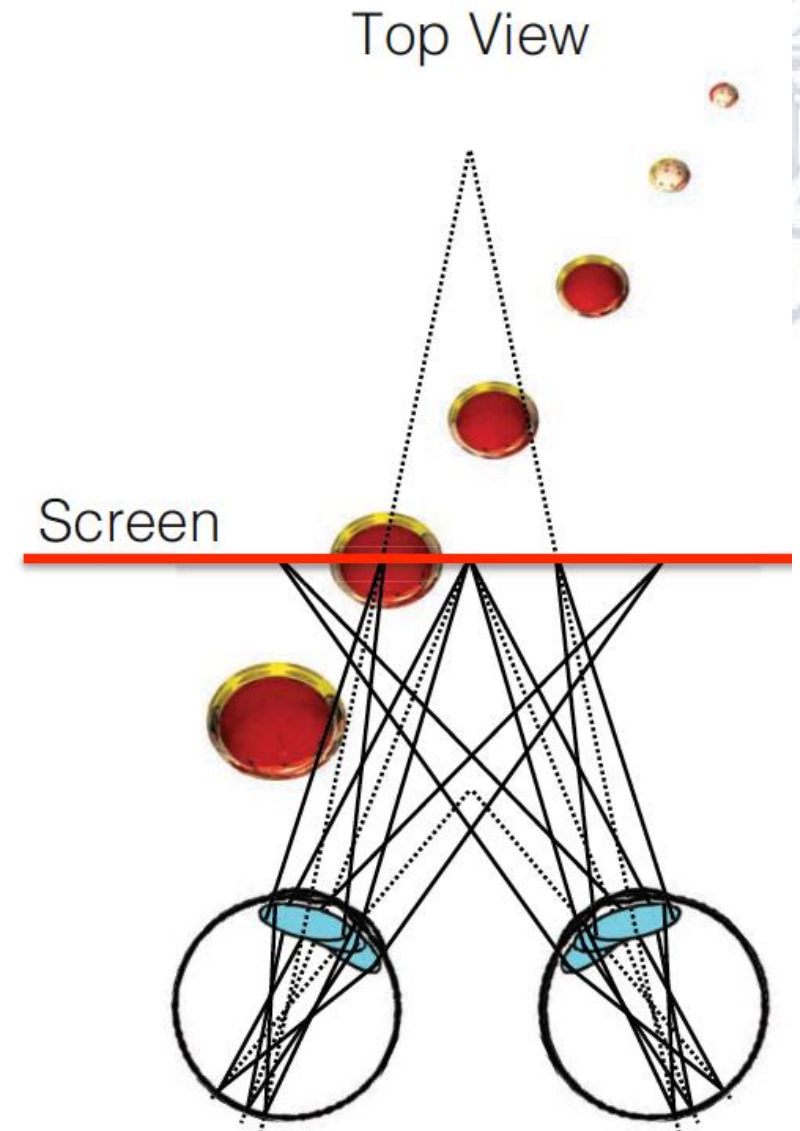
effects

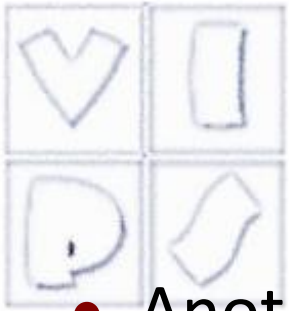
- visual discomfort
- visual fatigue
- nausea
- diplopic vision
- eyestrain
- compromised image quality
- pathologies in developing visual system
- ...



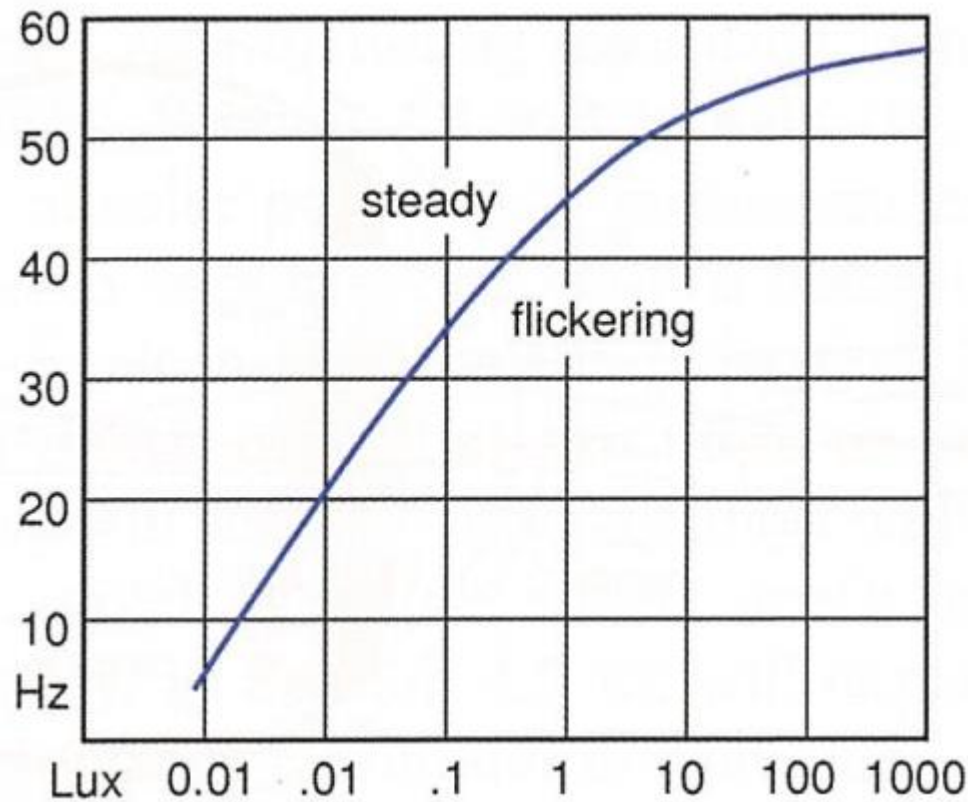
Stereo Displays Today:

Vergence-Accommodation **Mismatch!**





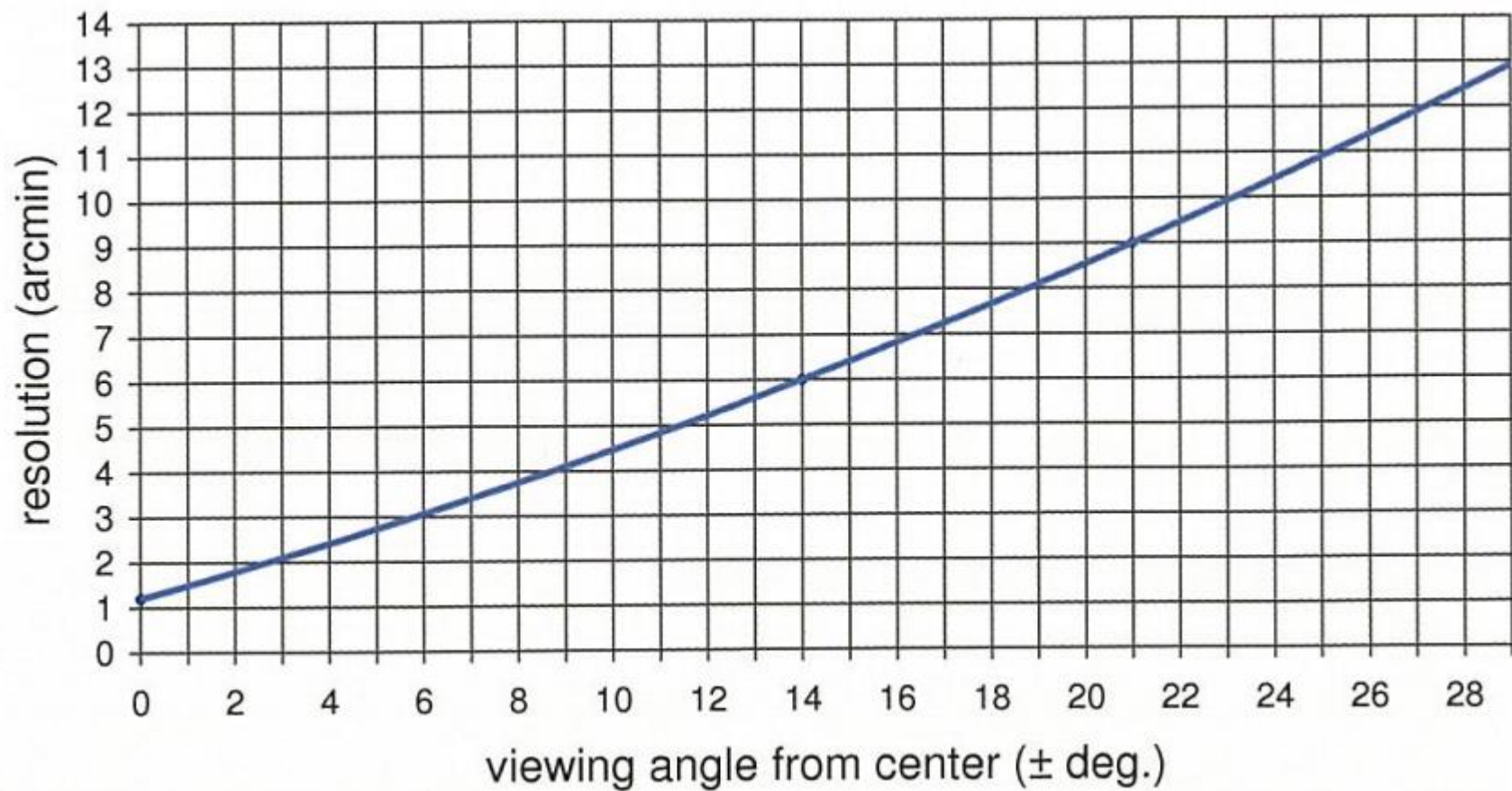
- Another issue for VR displays is Flicker Fusion Rate
 - <https://www.youtube.com/watch?v=fGGXEhOh-Is>





Resolution

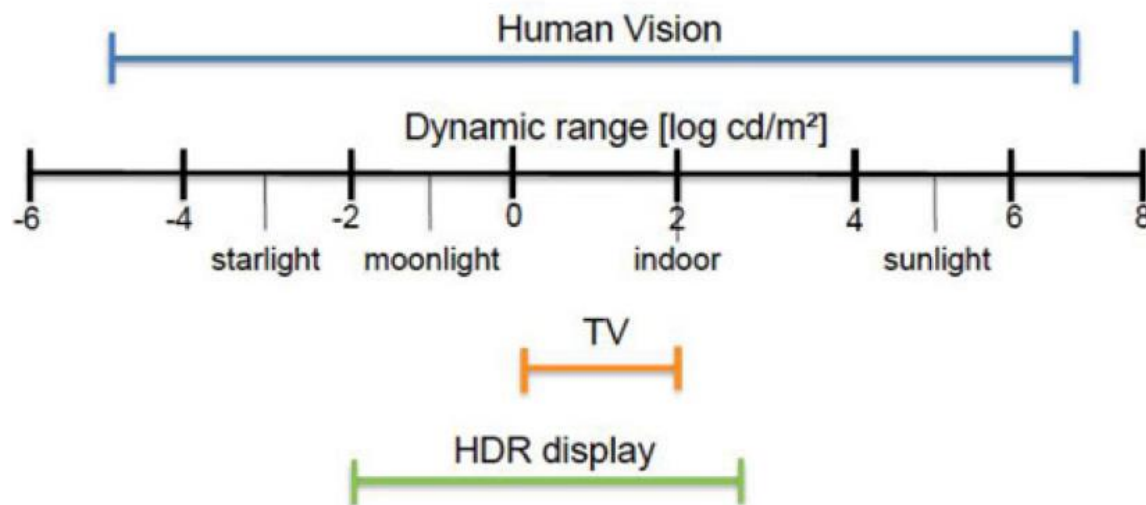
- Decrease far from fovea



Dynamic range

- Human vision has far higher dynamic range than any available display technology
 - 40 f-stops, cf 17 f-stops for HDR display

Human vision up to 12 decades/40f-stops



Source: <http://petapixel.com/>

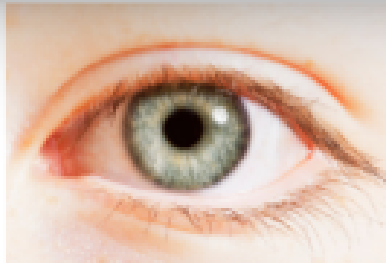


Human Vision System

- visual acuity: 20/20 is ~ 1 arc min
- field of view: $\sim 200^\circ$ monocular, $\sim 120^\circ$ binocular, $\sim 135^\circ$ vertical
- resolution of eye: ~ 576 megapixels
- temporal resolution: ~ 60 Hz (depends on contrast, luminance)
- dynamic range: instantaneous 6.5 f-stops, adapt to 46.5 f-stops
- colour: everything in CIE xy diagram
- depth cues in 3D displays: vergence, focus, (dis)comfort
- accommodation range: $\sim 8\text{cm}$ to ∞ , degrades with age

Can displays match?

- Not exactly...

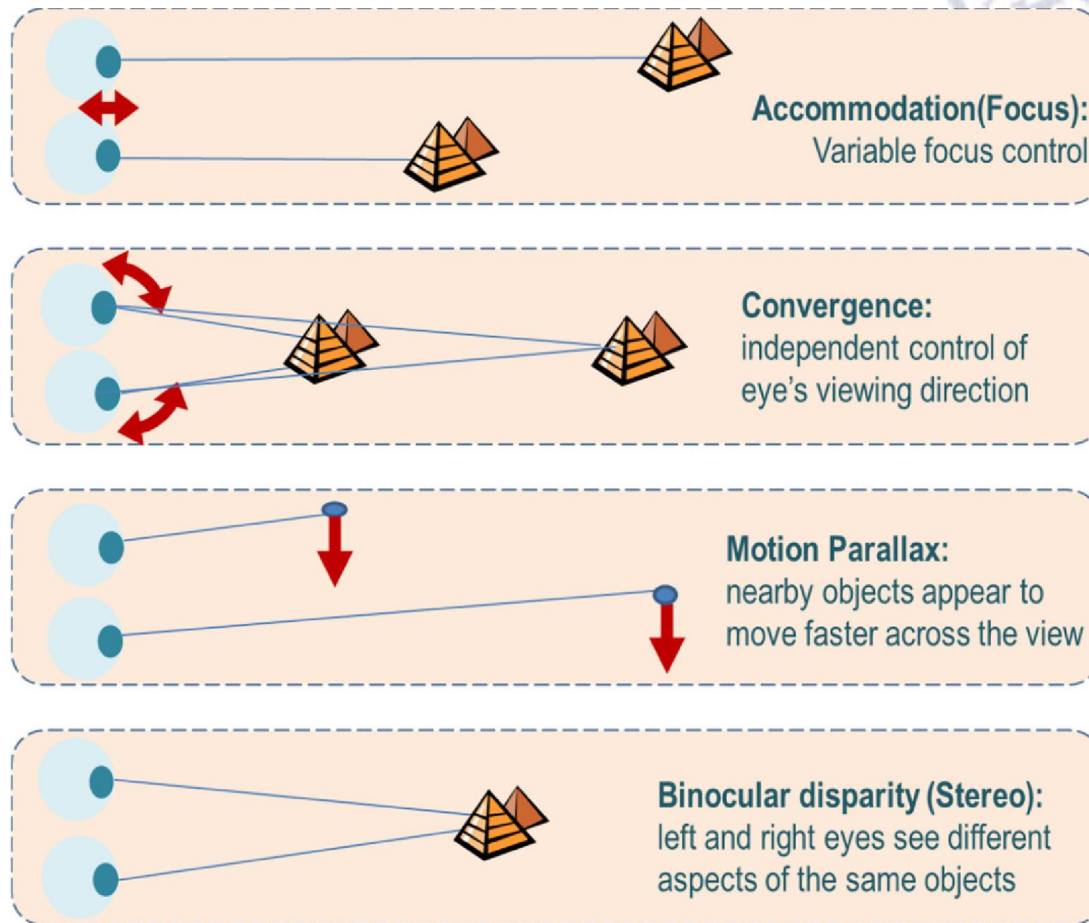


	Human Eyes	HTC Vive
FOV	200° x 135°	110° x 110°
Stereo Overlap	120°	110°
Resolution	30,000 x 20,000	2,160 x 1,200
Pixels/inch	>2190 (100mm to screen)	456
Update	60 Hz	90 Hz



Immersion: 3D cues

- Just stereo? No!



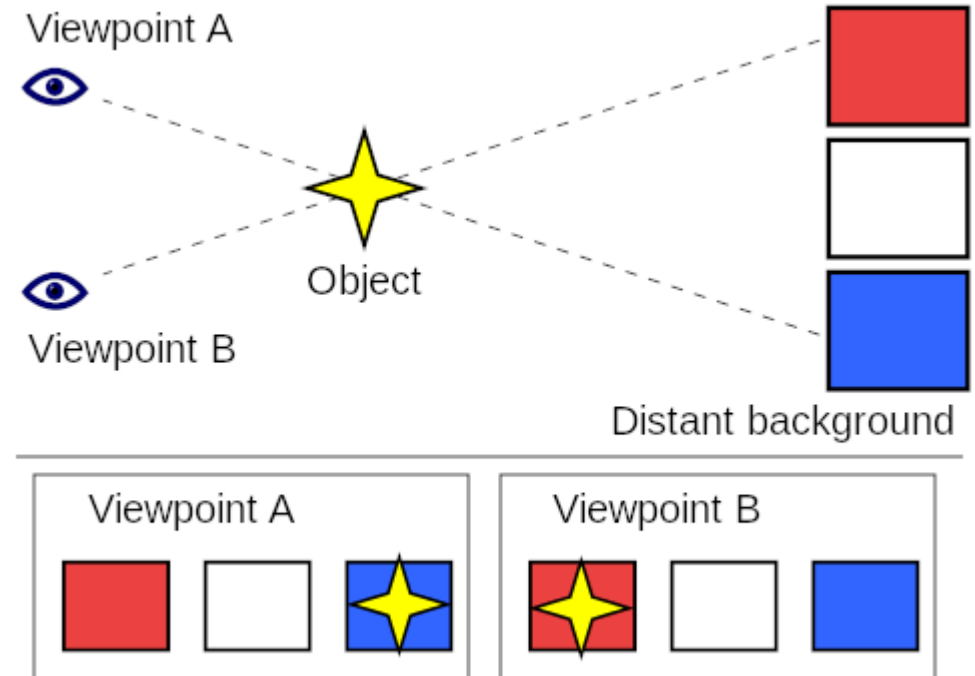
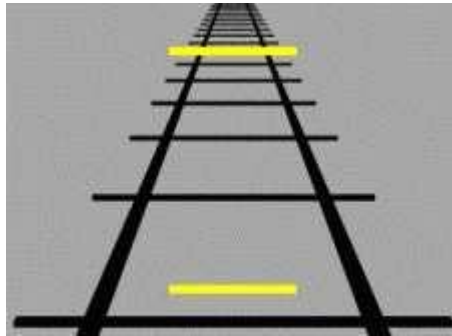
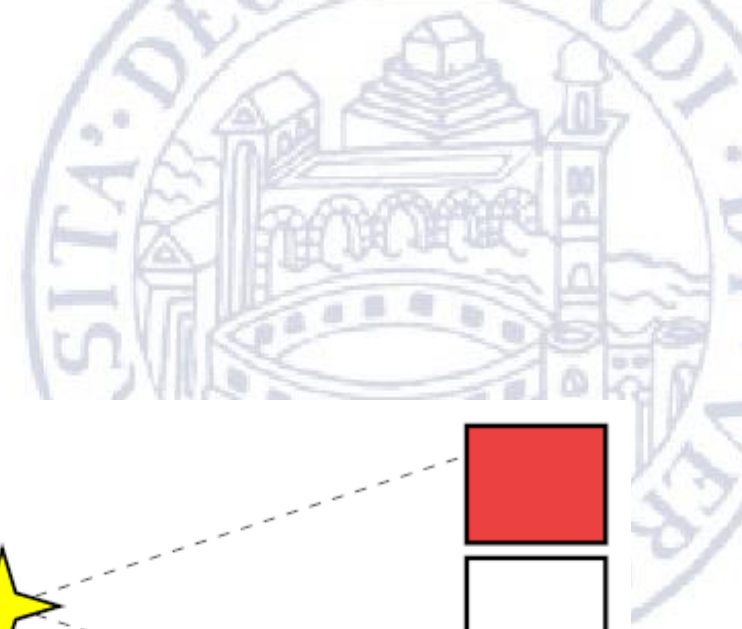


Immersion: 3D cues

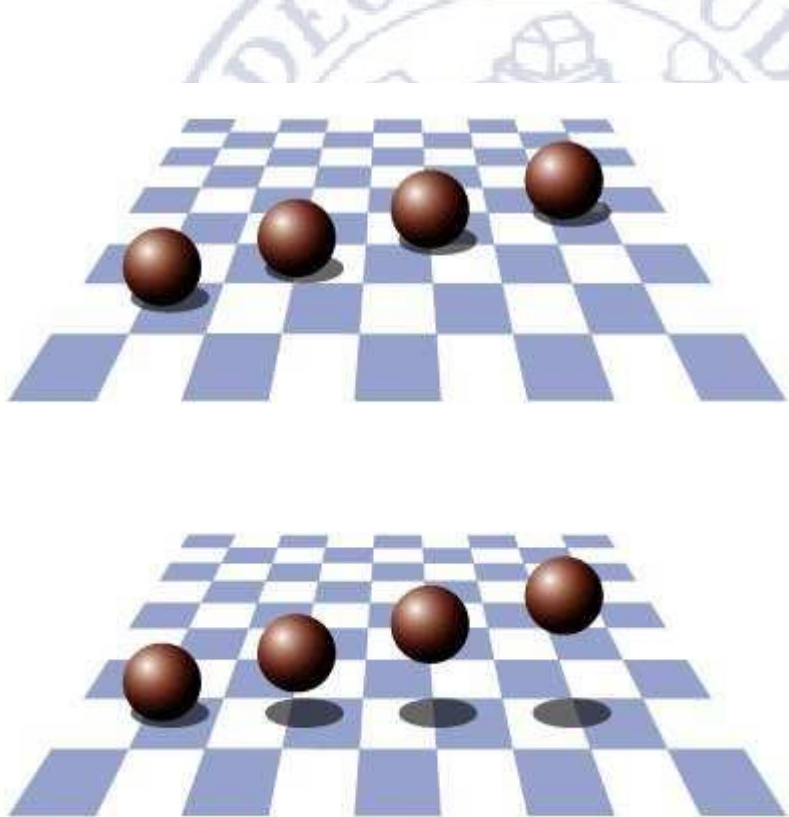
- Just stereo? No!
 - And also
 - atmospheric artifacts (fog)
 - relative sizes
 - image blur
 - occlusion
 - shadows
 - Texture
 - Colour – bluish objects more distant
 - Motion Dynamics – objects in relative motion

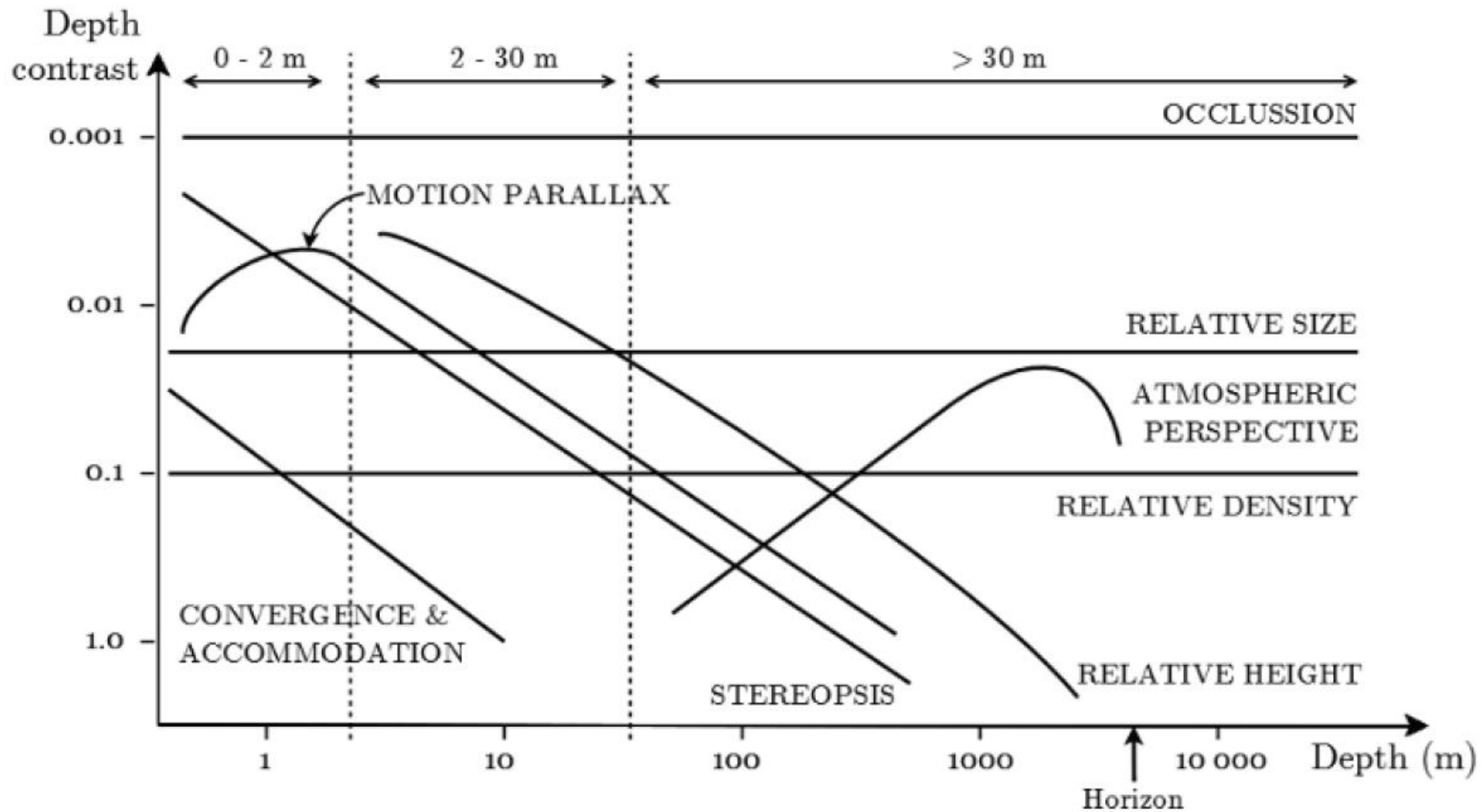
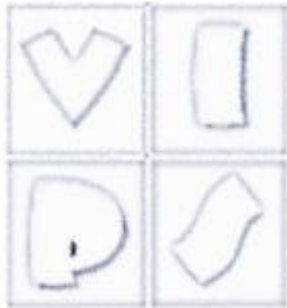






- Depth perception illusions
- <https://www.youtube.com/watch?v=p-eZcHPp7Go>







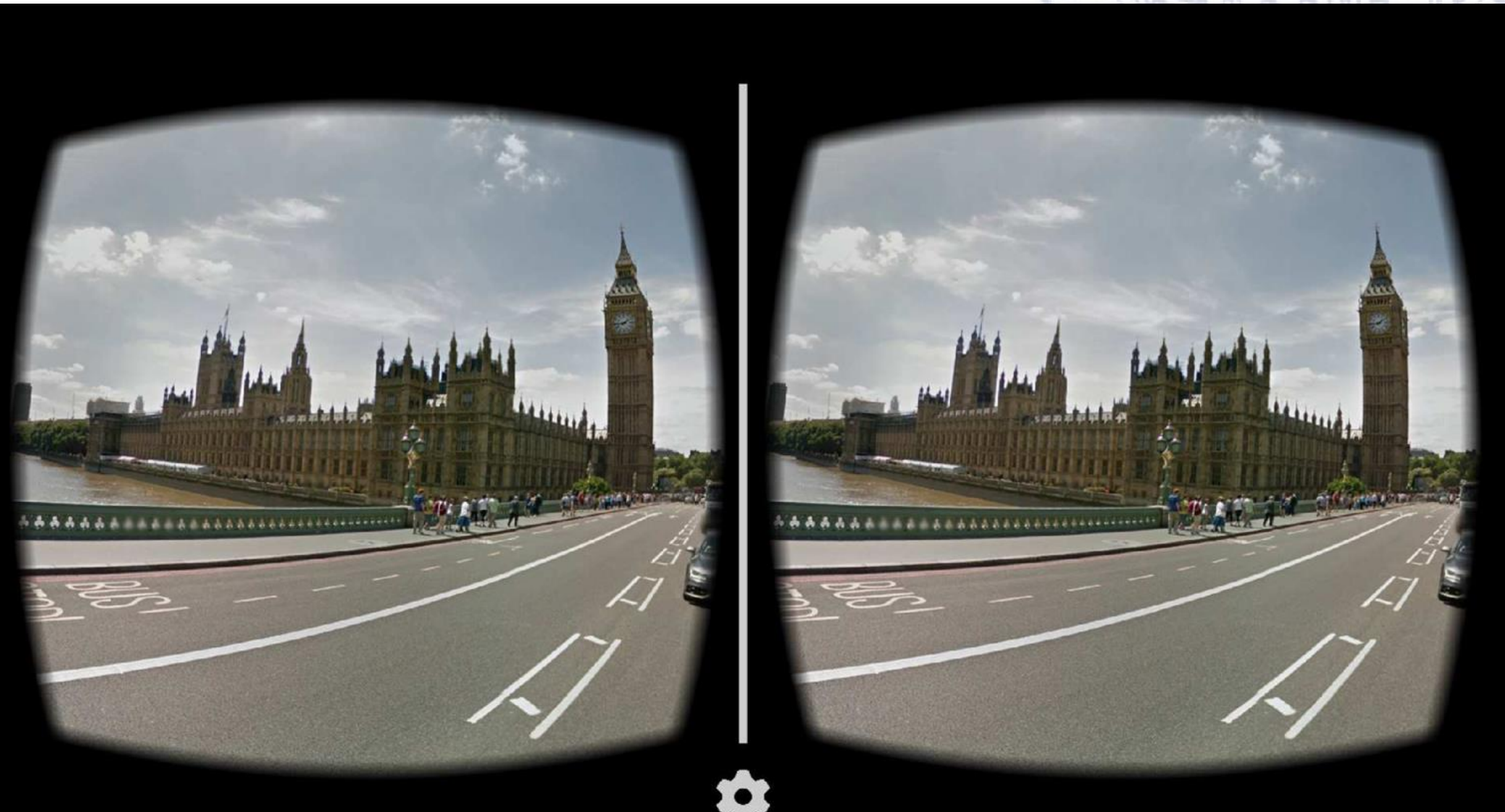
Issues for VR

- Coherence of depth cues?
 - Difficult
- Vergence-accommodation mismatch in VR headsets.
- Imperfect head tracking. If there is significant latency, then the visual stimuli will not appear in the correct place at the expected time.
- Many tracking systems track the head orientation only.



Street view in Google Cardboard

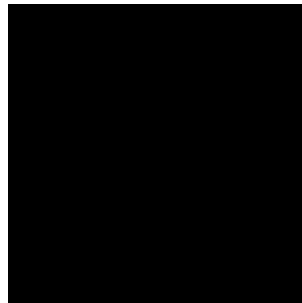
- No stereo cues. Still ok for users



Motion perception



- Motion detection depends on local processing and integrated later.
 - Optical flow
- Virtual Reality then uses apparent motion due to image sequences
 - Stroboscopic apparent motion
- Beta and Phi effects



- Persistence of Vision (100ms)?
 - Actually motion perceived also with 2fps?!?

Motion perception

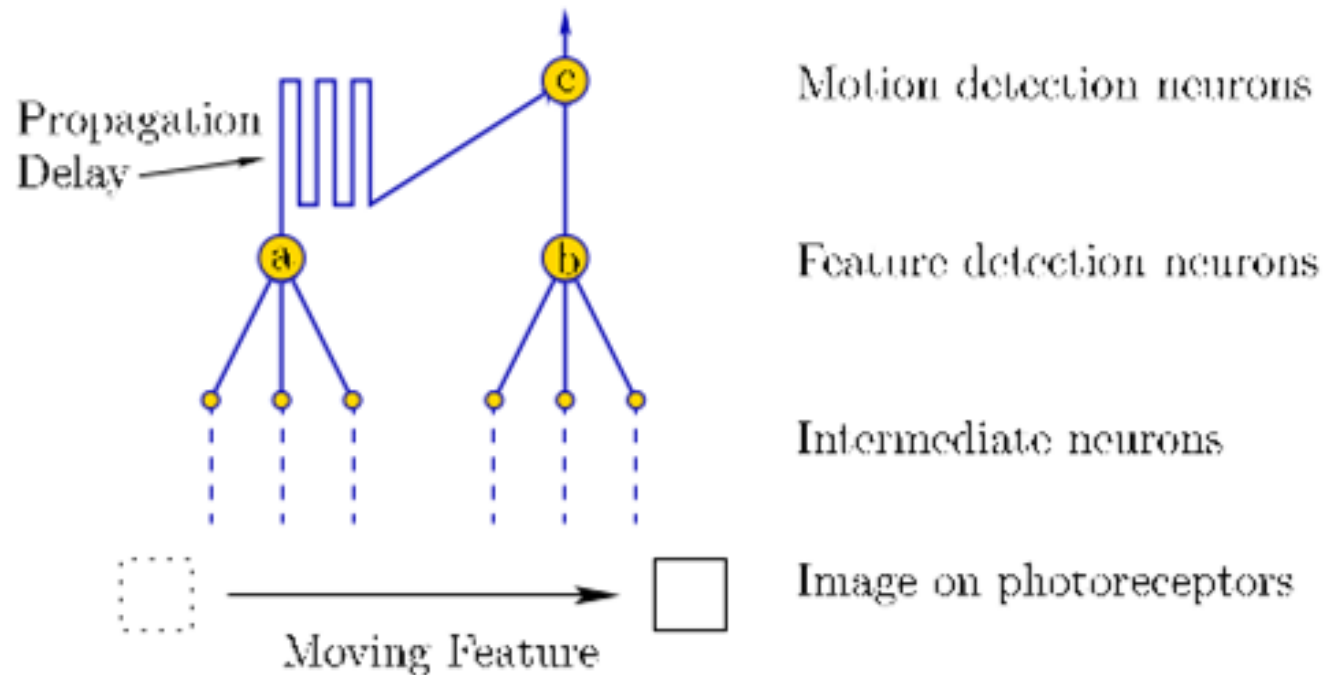
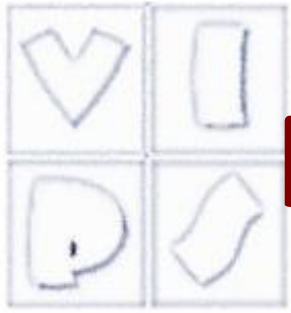


Figure 6.12: The neural circuitry directly supports motion detection. As the image feature moves across the retina, nearby feature detection neurons (labeled *a* and *b*) activate in succession. Their outputs connect to motion detection neurons (labeled



Frame rate and flickering

- 24 FPS may be enough to perceive motions smoothly,
- Due to black pauses in CRT
- In LCD caused by polarity inversion
- However there are other issues for VR displays (HMD)



Issues in HMD

- A solution may be reduce the time of image display
 - But flickering appears. Ok with frame rate > 90 fps
 - Fast pixel switch is needed. OLED < 0.1 ms

Virtual Object



Yawing Head
(top view)

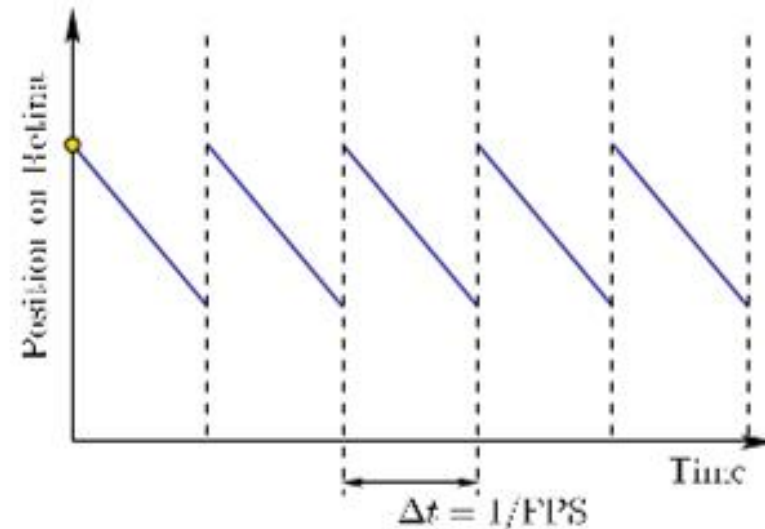


Figure 6.18: A problem with *perception of stationarity* under stroboscopic apparent motion: The image of a feature slips across the retina in a repeating pattern as the VOR is performed.



Other important senses in VR

- Haptic sensations
 - Somatosensory System
 - complex system of nerve cells that responds to changes to the surface or internal state of the body
 - Skin is the largest organ 1.3-1.7 square m in adults
 - Tactile: Surface properties
 - Receptors not evenly spread
 - Most densely populated area is the tongue
- Kinesthetic: Muscles, Tendons, etc.
 - Also known as proprioception

Other important senses in VR

- Haptic sensations

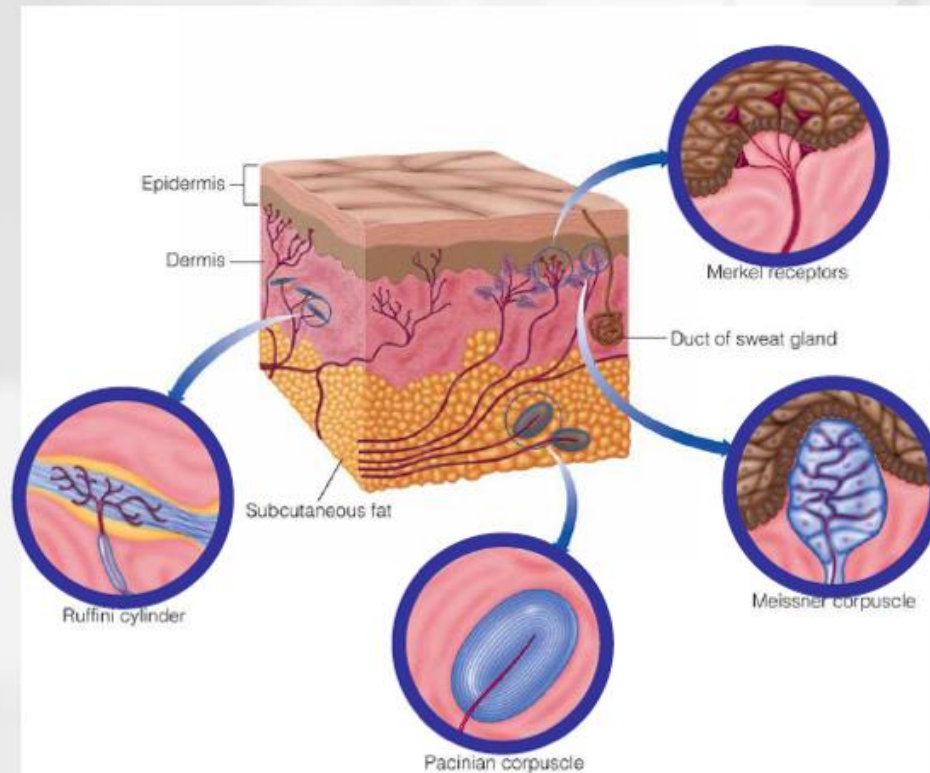
Mechanoreceptors

Merkel receptor - disk-shaped receptor located near the border between the epidermis and dermis

Meissner corpuscle - stack of flattened disks in the dermis just below epidermis

Ruffini cylinder - branched fibers inside a cylindrical capsule

Pacinian corpuscle - onion-like capsule located deep in the skin





Other important senses in VR

- Mechanoreceptors
 - Exploited by VR haptic feedback devices
 - Cells that respond to pressure, stretching, and vibration
 - Slow Acting (SA), Rapidly Acting (RA)
 - Type I at surface – light discriminate touch
 - Type II deep in dermis – heavy and continuous touch

Receptor Type	Rate of Acting	Stimulus Frequency	Receptive Field	Detection Function
Merkel Discs	SA-I	0 – 10 Hz	Small, well defined	Edges, intensity
Ruffini corpuscles	SA-II	0 – 10 Hz	Large, indistinct	Static force, skin stretch
Meissner corpuscles	RA-I	20 – 50 Hz	Small, well defined	Velocity, edges
Pacinian corpuscles	RA-II	100 – 300 Hz	Large, indistinct	Acceleration, vibration



Touch sensitivity

- Two point test



Body Site Threshold Distance

- Finger 2-3mm
- Cheek 6mm
- Nose 7mm
- Palm 10mm
- Forehead 15mm
- Foot 20mm
- Belly 30mm
- Forearm 35mm
- Upper Arm 39mm
- Back 39mm
- Shoulder 41mm
- Thigh 42mm
- Calf 45mm



Other important senses in VR

- Proprioception (joint position sense)
 - Awareness of movement and positions of body parts
 - Due to nerve endings and Pacinian and Ruffini corpuscles at joints
 - Enables us to touch nose with eyes closed
 - Joints closer to body more accurately sensed
 - Users know hand position accurate to 8cm without looking at them
- Kinaesthesia (joint movement sense)
 - Sensing muscle contraction or stretching
 - Cutaneous mechanoreceptors measuring skin stretching
- Helps with force sensation



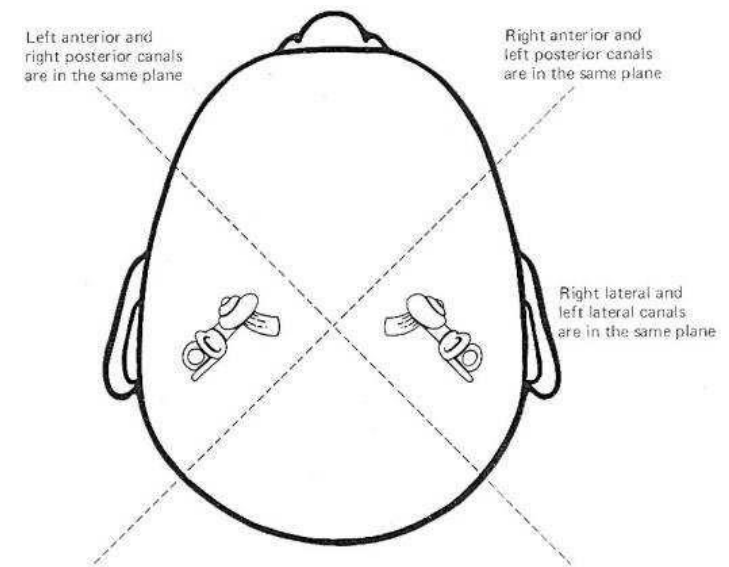
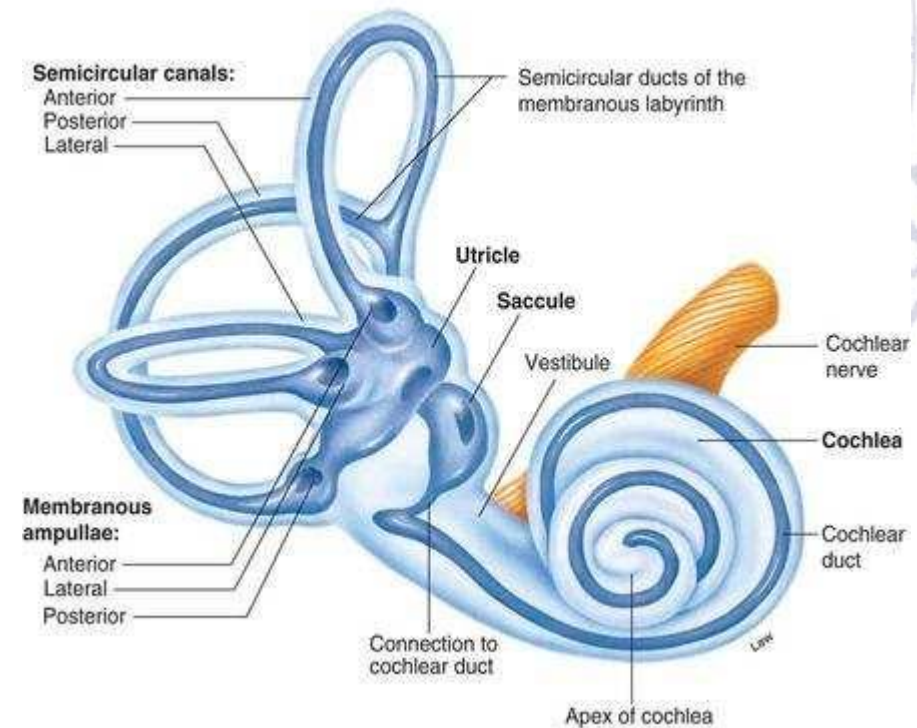


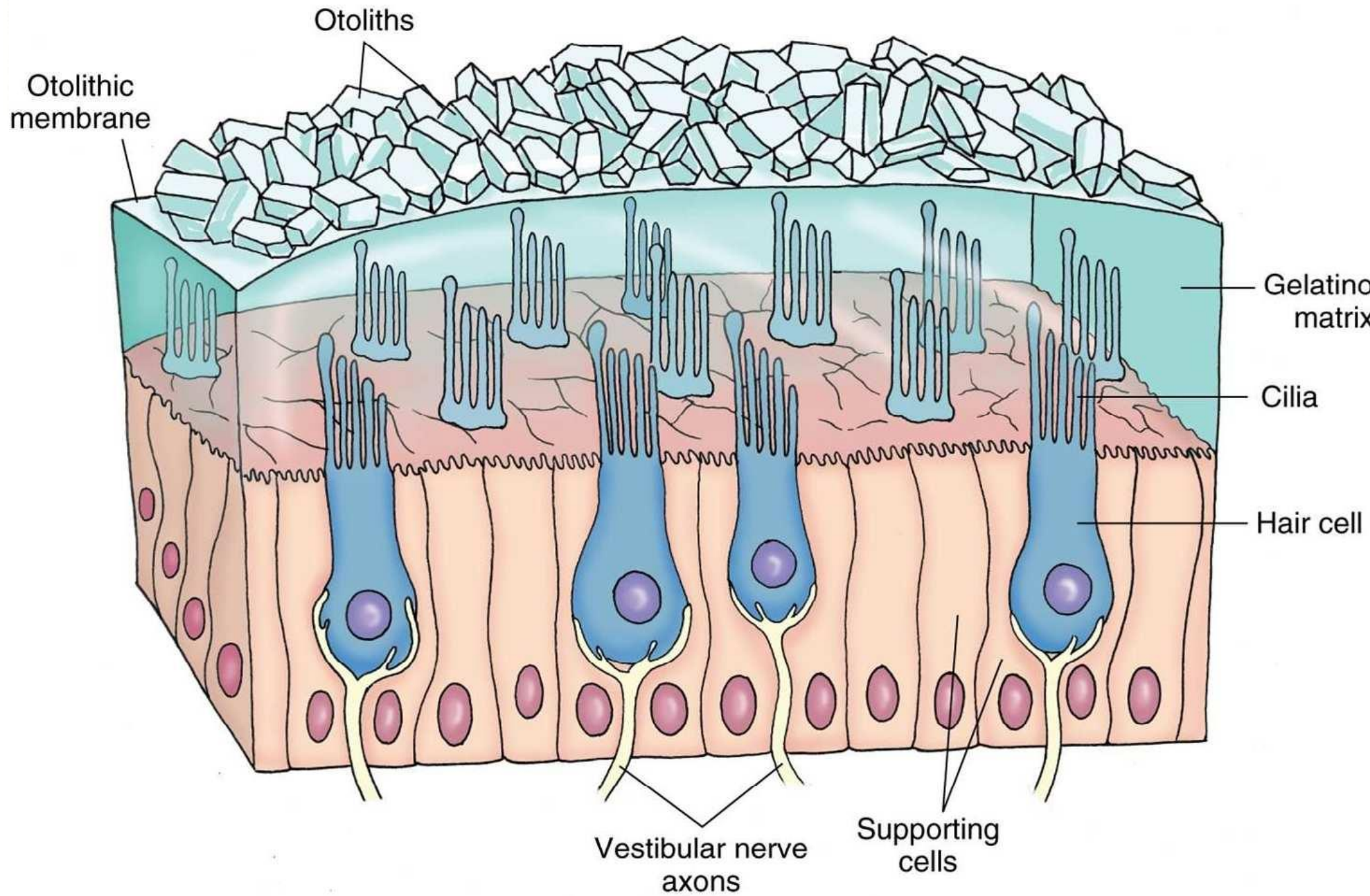
Proprioception

- The motor cortex, which controls body motion, sends signals called **efference copies to other parts of the brain to communicate what motions have been executed**
- It corresponds, in robots, to having encoders on joints or wheels, to indicate how far they have moved.
- Implication: you cannot tickle yourself because you know where your fingers are moving;

Vestibular system

- The cochlea handles hearing, the remaining parts belong to the vestibular system.
- The utricle and saccule measure linear acceleration; together they form the otolith system.
- The semicircular canals measure angular acceleration.







Audio

- In the visual case, we saw that perceptual experiences are often surprising because they are based on adaptation, missing data, assumptions filled in by neural structures, and many other factors. The same is true for auditory experiences
 - Many interesting aspects are derived from psychophysical measurements

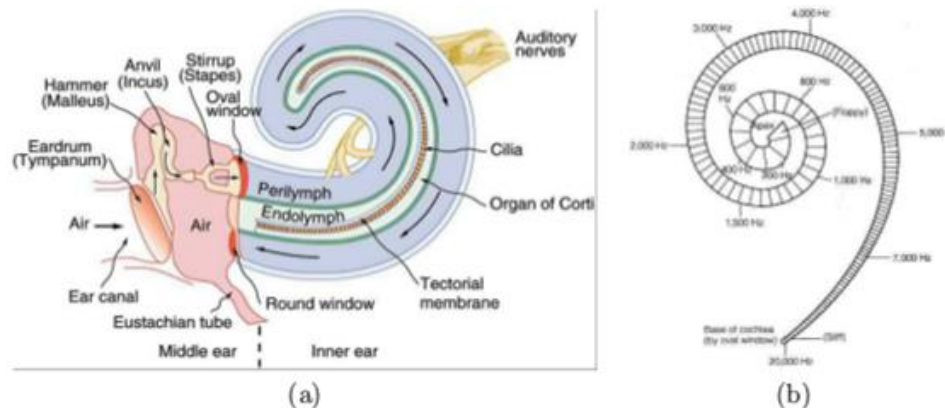


Figure 11.5: The operation of the cochlea: (a) The perilymph transmits waves that are forced by the oval window through a tube that extends the length of the cochlea and back again, to the round window. (b) Because of varying thickness and stiffness, the central spine (basilar membrane) is sensitive to particular frequencies of vibration, thus providing the mechanism for analyzing sound by frequency.



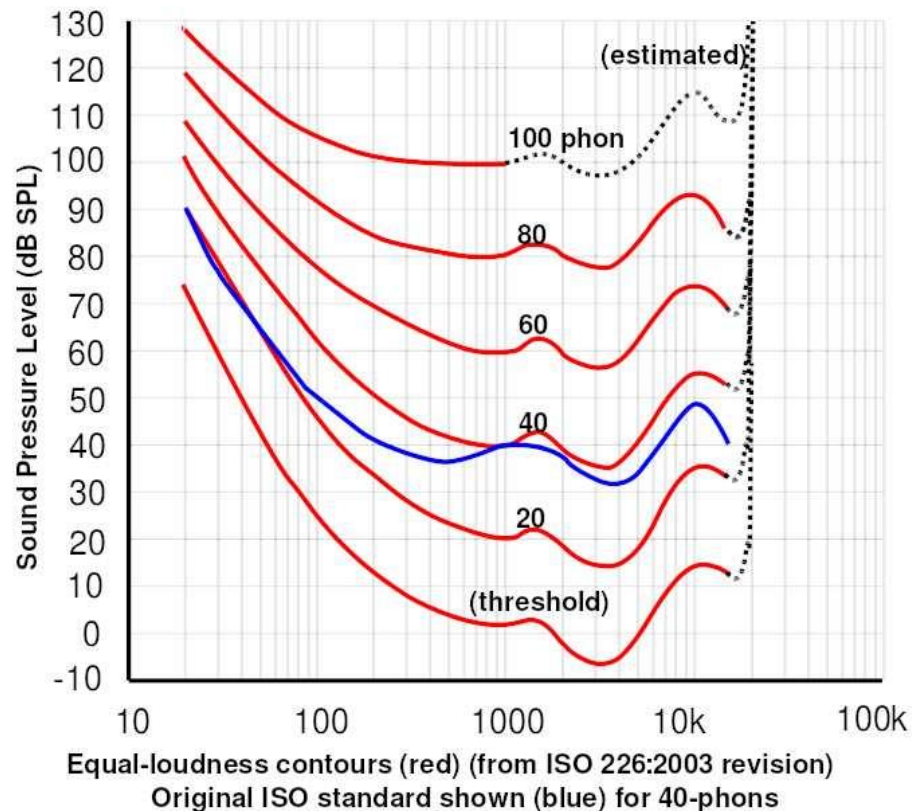
Auditory illusions

- Precedence effect,
 - one sound is perceived if two nearly identical sounds arrive at slightly different times. Sounds often reflect from surfaces, causing reverberation, which is the delayed arrival at the ears of many “copies” of the sound. Rather than hearing a jumble, people perceive a single sound. This is based on the first arrival, which usually has the largest amplitude.
- An echo is perceived if the timing difference is larger than the echo threshold



Loudness perception

- Variable with frequency
 - To be considered in experiments with audio interaction





Steven's law

- One thing to be considered is that the response to a stimulus is nonlinear and depends on the stimulus type

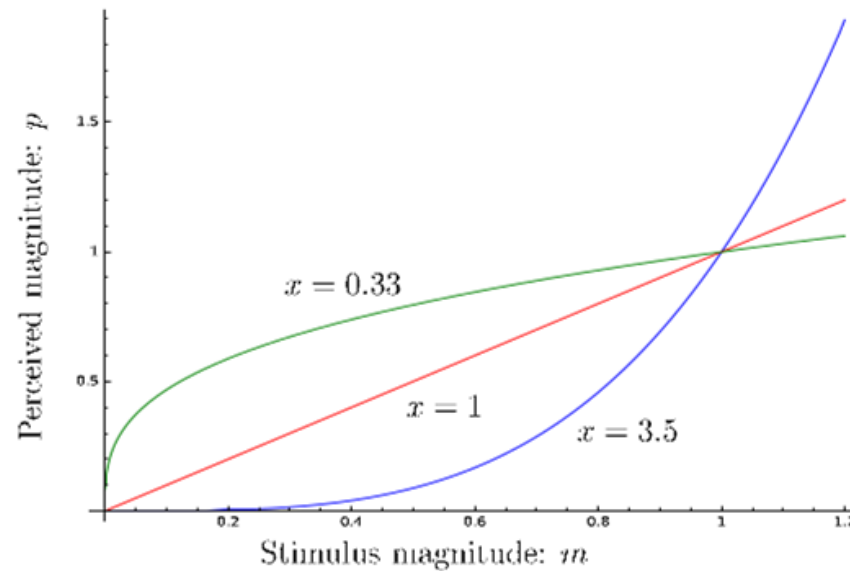


Figure 2.22: Steven's power law (2.1) captures the relationship between the magnitude of a stimulus and its perceived magnitude. The model is an exponential curve, and the exponent depends on the stimulus type.



JND

- Another key psychophysical concept is the just noticeable difference (JND). This is the amount that the stimulus needs to be changed so that subjects would perceive it to have changed in at least 50 percent of trials.
- For a large change, all or nearly all subjects would report a change. If the change is too small, then none or nearly none of the subjects would notice.
- The experimental challenge is to vary the amount of change until the chance of someone reporting a change is 50 percent.



Sound localization

- Estimating the location of a sound source by hearing it.
 - crucial for many VR experiences. Auditory and visual cues should match. For example, if people are socializing, then their voices should seem to come from the mouths of corresponding avatars.
- The JND concept is applied for localization to obtain the minimum audible angle (MAA), the minimum amount of angular variation that can be detected by a human listener.
- A spherical coordinate (azimuth, elevation) system is usually used for localization, in which the listener's head is at the origin;

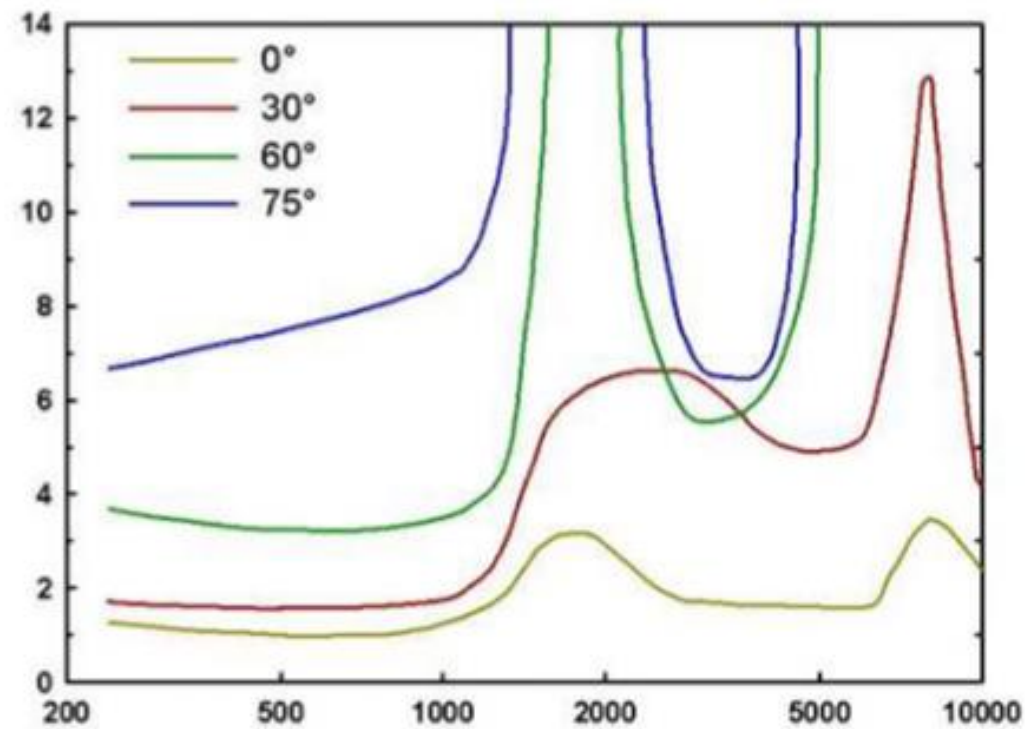
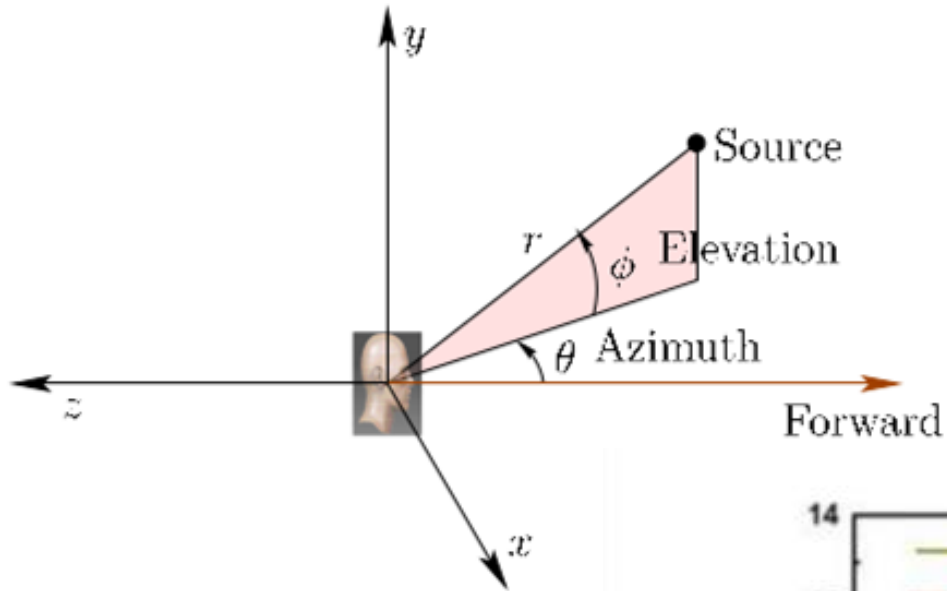


Figure 11.10: Plots of the *minimum audible angle* (MAA) as a function of frequency. Each plot corresponds to a different azimuth angle.



Monaural cues for sound localization

- The pinna is shaped asymmetrically so that incoming sound is distorted in a way that depends on the direction
- The amplitude of a sound decreases quadratically with distance. If it is a familiar sound, then its distance can be estimated from the perceived amplitude.
- For distant sounds, a distortion of the frequency spectrum occurs because higher-frequency components attenuate more quickly than low-frequency components.
 - For example, distant thunder is perceived as a deep rumble, but nearby thunder includes a higher-pitched popping sound
- Reverberation can be used. This cue alone is called echolocation, which is used naturally by some animals, including bats. Some people can perform this by making clicking sounds or other sharp noises; this allows acoustic wayfinding for blind people.



Binaural cues

- Interaural level difference (ILD), difference in sound magnitude as heard by each ear.
- Interaural time difference (ITD),
 - The distance between the two ears is approximately 21.5cm, which results in different arrival times of the sound from a source. Note that sound travels 21.5cm in about 0.6ms, which means that small differences are used for localization.

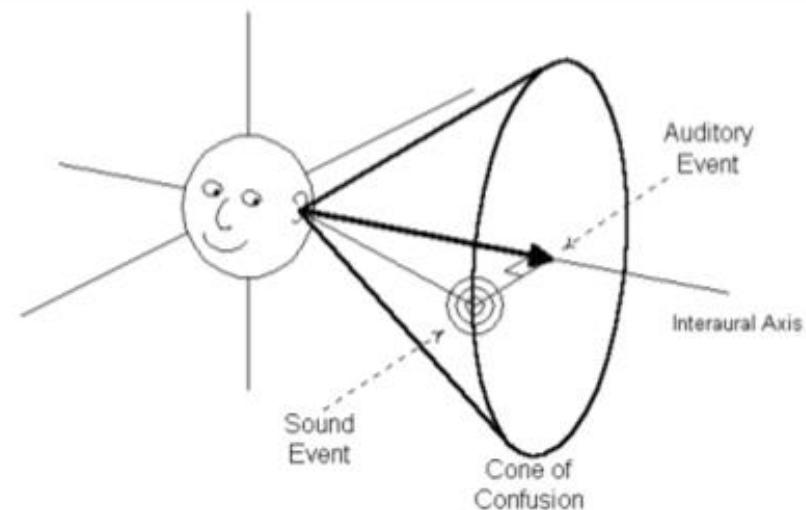


Figure 11.11: The *cone of confusion* is the set of locations where a point source



Hierarchical processing

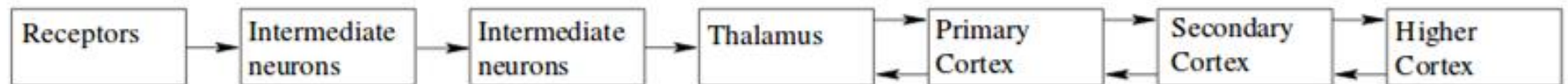
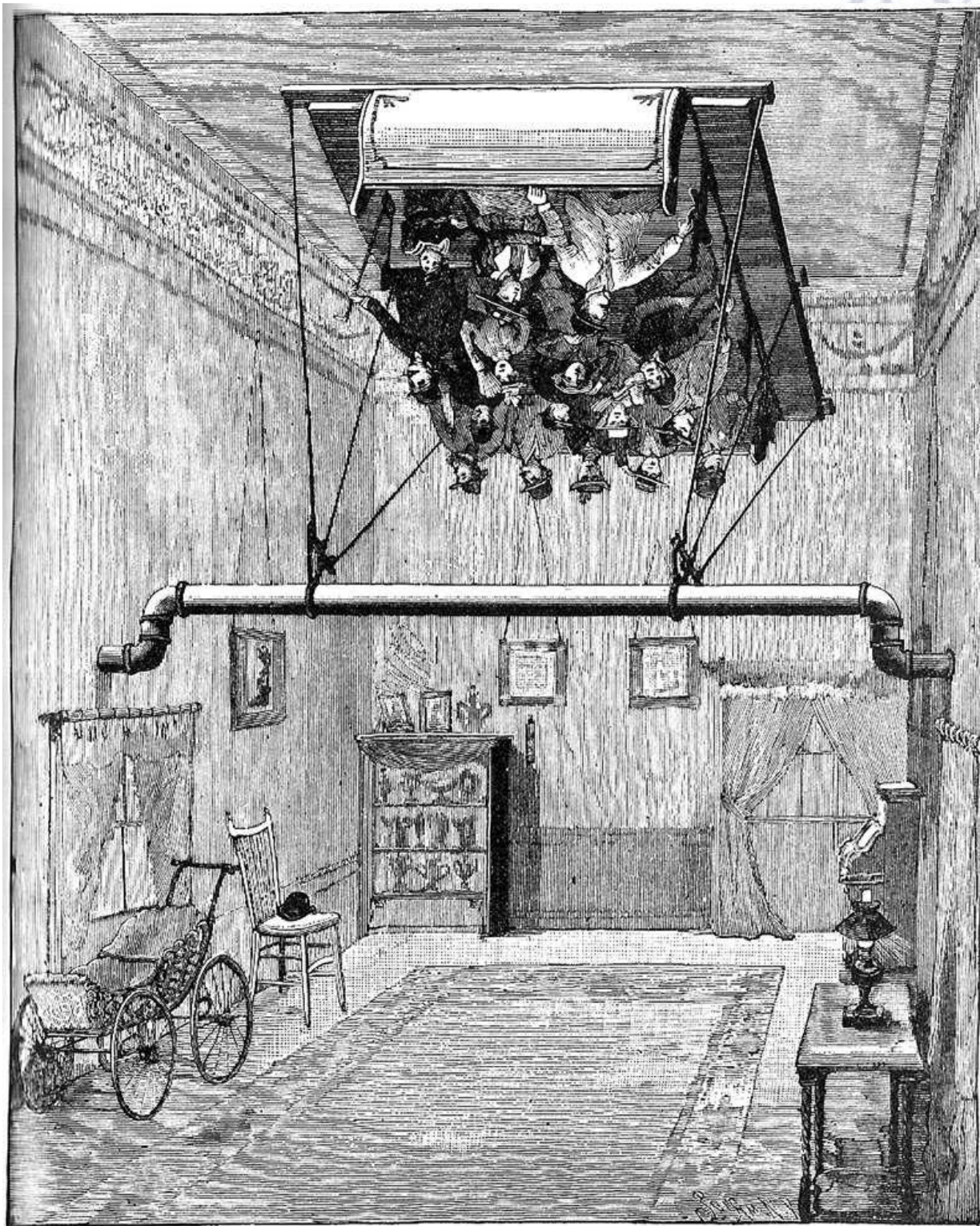


Figure 2.19: The stimulus captured by receptors works its way through a hierarchical network of neurons. In the early stages, signals are combined from multiple receptors and propagated upward. At later stages, information flows bidirectionally.



Information fusion

- Signals from multiple senses and proprioception are being processed and combined with our experiences by our brain
- We expect coherent information
- VR often cannot provide all the related sensations related to a simulation
 - This may cause problems
 - We could develop symptoms of dizziness or nausea
- Example:vection, e.g. illusion of self motion. Vision sense reports to your brain that you are accelerating but your balance sense reports that you are motionless.



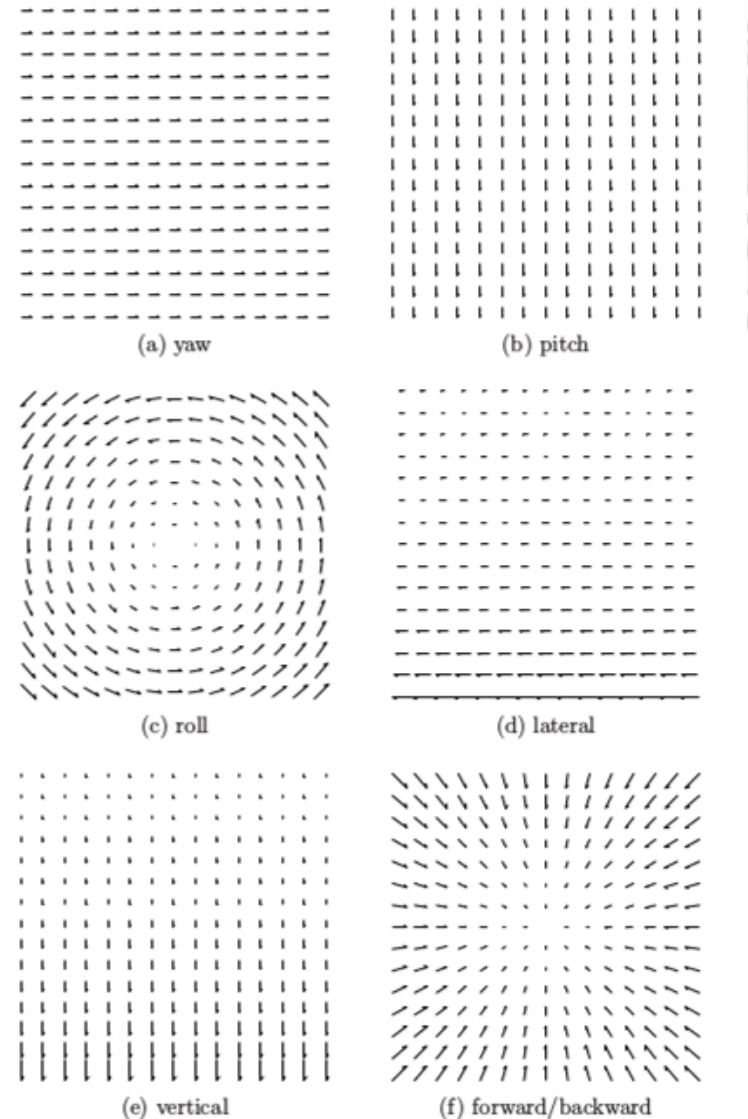
ILLUSION PRODUCED BY A RIDE IN THE SWING.

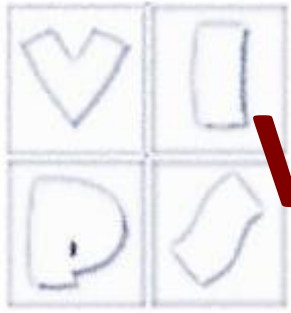




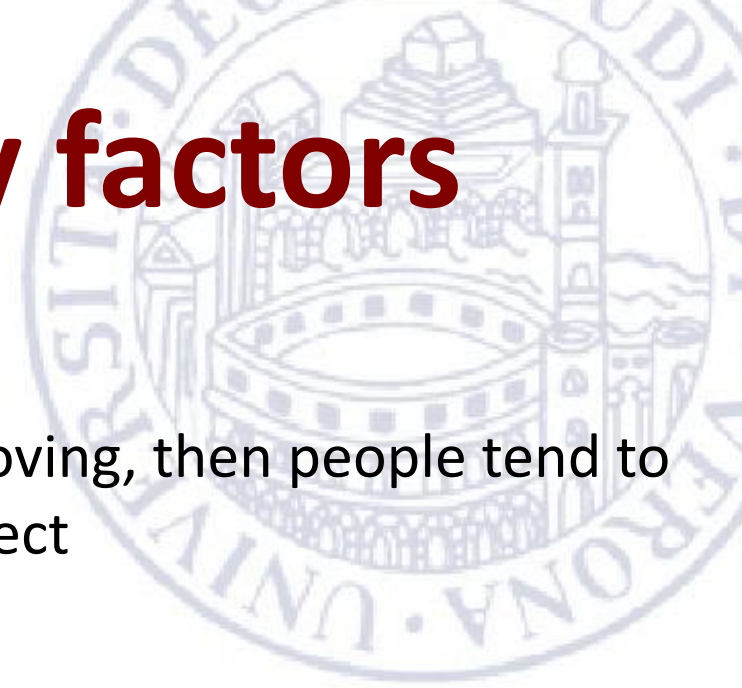
Optical flow

- Motion is perceived (like in artificial systems) by estimation of “optical flow” from image sequences
- Flow vectors are used to recover motion type
- Different types of vection (motion perception from flow)

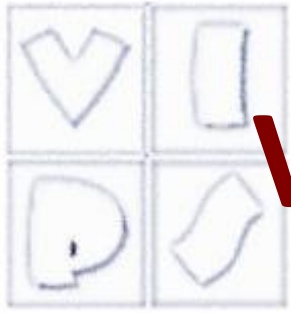




Vection sensitivity factors



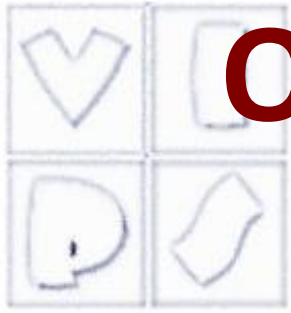
- Percentage of field of view:
 - If only a small part of the visual field is moving, then people tend to perceive that it is caused by a moving object
- Distance from center view
- Exposure time:
 - The perception of self motion due to vection increases with the time of exposure. In a few milliseconds, no vection may occur.
- Spatial frequency:
 - Optical flow perception is enhanced by detail
- Contrast:
 - Contrast also increase optical flow signal
- Other sensory cues
 - If coherent with real/false motion can increase the correct/incorrect perception (e.g. noise, etc.)



Vection sensitivity factors



- Prior knowledge:
 - Knowing in advance what is going to happen, our perception may be changed
- Attention:
 - If the user is distracted by another activity, such as aiming a virtual weapon or selecting a menu option, then vection and its side effects may be mitigated.
- Prior training or adaptation:
 - With enough exposure, the body may learn to distinguish vection from true motion to the point that vection becomes comfortable. Thus, many users can be trained to overcome VR sickness
 - through repeated, prolonged exposure.



Other sensor fusion effect: McGurk effect

- Not relevant for VR
- The McGurk effect is an experiment that clearly indicates the power of integration by mixing visual and auditory cues [207]. A video of a person speaking is shown with the audio track dubbed so that the spoken sounds do not match the video. Two types of illusions were then observed:
 - If “ba” is heard and “ga” is shown, then most subjects perceive “da” being said. This corresponds to a plausible fusion of sounds that explains the mismatch, but does not correspond to either original cue. Alternatively, the sounds may combine to produce a perceived “bga” in the case of “ga” on the sound track and “ba” on the visual track.



<https://www.youtube.com/watch?v=G-IN8vWm3m0>



References

- VRBook chapters 2, 6, 8

